

Thermal Vias Modelling in PCB for Simulation

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Abstract

Paper elaborates about the various thermal vias modelling approaches commonly used. Extensive simulation studies were conducted in both natural and forced cooling methods with different vias array sizes. The investigation also includes the simulation results of all vias models, discusses on their accuracies and merits in thermal analysis.

1. Overview of Thermal Vias & Different Modelling Approaches

Thermal vias are plated through holes (PTHs) that are provided below the heat-source components in PCB. These vias establish a significant heat path from the top copper layer to the bottom side of the PCB through lowering the through-plane thermal resistance of PCB under the component.

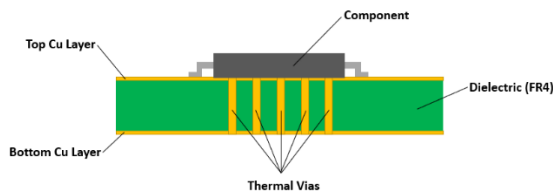


Figure 1: Thermal Vias in PCB

Following parameters play an important role for effective heat dissipation through the vias:

- No of vias & its array pattern based on heat-source size
- Via pitch
- Plating thickness

In general, construction of geometrically exact thermal vias requires the fine local mesh & can take up a high computation space in simulation, especially when a design has thermal vias solution for many components. So, knowing other simplified modelling approaches becomes essential & helps the thermal engineers in predicting the component temperature with same level of accuracy as with exact model.

1.1 Exact Via Model

Exact thermal vias are geometrically exact & a via-array of this model is shown below.

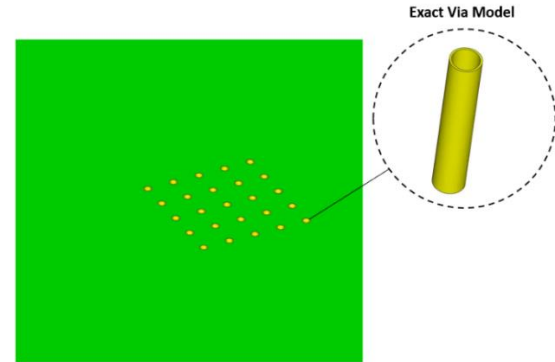


Figure 2: Exact Via Model

As the geometrically thin feature of copper plating thickness is modelled, it demands a fine local mesh.

1.2 Lumped individual vias model

In this approach, each via is individually lumped and an equivalent thermal conductivity assigned to it. Effective thermal conductivity of each via can be calculated using the volume averaged-method formula as given below.

Effective Thermal Conductivity:

$$K_e = K_{Cu} \times \alpha + K_{air} \times (1 - \alpha)$$

where

K_{Cu} : Thermal Conductivity of Copper

K_{air} : Thermal Conductivity of Air

α : Fraction of Copper Content in a via

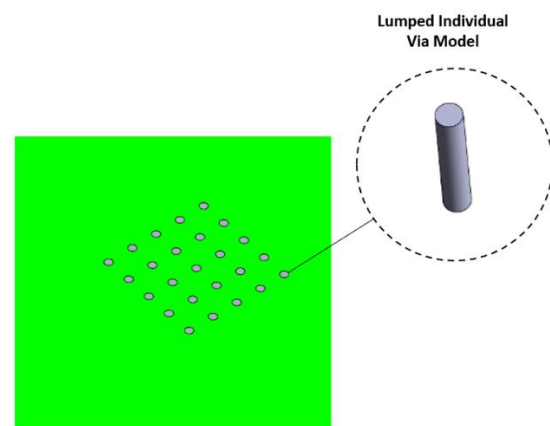


Figure 3: Individually Lumped Via Model

As the copper plating thickness is not included in this modelling approach, fine local meshing is avoided, but still this model could become computational expensive in a system-level analysis.

1.3 Lumped Vias Array Model

It is widely used modelling approach for thermal via array, a single lumped block is modelled as same size of vias array, and its thickness is equal to the thickness of PCB as shown below.

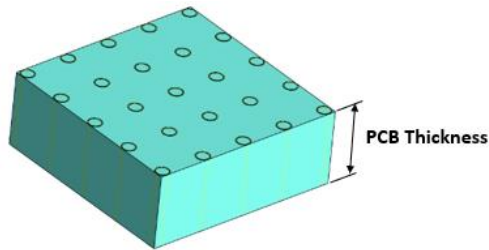


Figure 4: Lumped Vias Array Model

Biaxial conductivity of “In-Plane conductivity” & “Through-Plane conductivity” assigned to this block as follows:

$K_{\text{In-Plane of Via Array Block}} = K_{\text{In-Plane of PCB}}$

$K_{\text{Through-Plane of Via Array Block}}$ is calculated using the volume averaged-method formula:

$$K_{\text{Cu}} \times \alpha_{\text{Cu}} + K_{\text{air}} \times \alpha_{\text{air}} + K_{\text{FR4}} \times (1 - \alpha_{\text{Cu}} - \alpha_{\text{air}})$$

where

K_{Cu} : Thermal Conductivity of Copper

K_{air} : Thermal Conductivity of Air

K_{FR4} : Thermal Conductivity of FR4

α_{Cu} : Fraction of Copper in Via Array Block

α_{air} : Fraction of Air in Via Array Block

1.4 Lumped Component Size Model

This model is quick to build & does not require the via array placement information. It is similar to the “Lumped Vias Array Model”, but in this approach, instead of vias array size, the component size is considered for volume-averaged lumped method. The modelling lump is shown below.

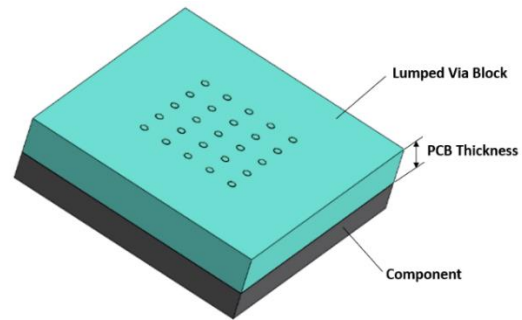


Figure 5: Lumped Component Size Model

In addition to the above modelling approaches, without vias & including only vias holes in PCB models are also investigated in this study using the CFD simulation methodology.

The simulation setup, conditions & results of different cases are presented in the following section.

2. Simulation Setup & Boundary Conditions

Simulation model, Setup & Boundary conditions considered are described below:

- Two signal & two plane (2S2P) layered PCB of size 80.0 x 50.0 x 1.6 mm. Each signal layer is 70 μ m thickness with 30% copper coverage and each plane layer is 35 μ m thickness with 80% copper coverage and dielectric material is taken as FR4.
- PCB is modelled with Orthotropic thermal conductivity approach.
- IC package size of 10.0 x 10.0 x 1.2 mm with exposed pad of size 4.5 x 4.5 mm dissipating 1.5 Watt located at the centre of the PCB.
- Ambient temperature of 30°C is applied.
- Both Natural cooling & forced cooling methods are simulated. For forced cooling analysis, air velocity of 2.0 m/s is included. Refer the figure 6.
- Via array patterns of 5 x 5, 4 x 4, 3 x 3 & 2 x 2 with 0.3mm via hole diameter, plating thickness 0.02mm & pitch 1.0mm are used.

The following cases are studied for both natural and forced cooling methods and the simulation results of all modelling approaches are provided in this section.

No Vias	Vias Holes	Exact Vias	Inv Via Lump	Via Array Lump	Comp Size Lump
	5 x 5	5 x 5	5 x 5	5 x 5	5 x 5
	4 x 4	4 x 4	4 x 4	4 x 4	4 x 4
	3 x 3	3 x 3	3 x 3	3 x 3	3 x 3
	2 x 2	2 x 2	2 x 2	2 x 2	2 x 2

Table 1: Simulation Cases

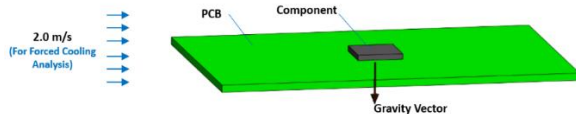


Figure 6: Simulation Model

3. Simulation Results

3.1 Natural Cooling Simulation Analysis

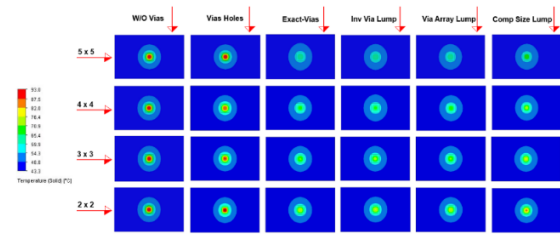


Figure 7: Temperature Contours of all modelling approaches

Array Size	Modelling Approach					
	W/O Vias	Via Holes	Exact Vias	Inv Array Lump	Via Array Lump	Comp Size Lump
	Component Temperature (°C)					
5 x 5	93.0	89.0	64.1	65.2	64.3	70.4
4 x 4		89.0	67.0	67.7	68.9	73.7
3 x 3		90.0	71.7	72.7	72.3	78.9
2 x 2		90.7	77.1	77.7	77.6	84.8

Table 2: Simulated Component Temperatures of all modelling approaches

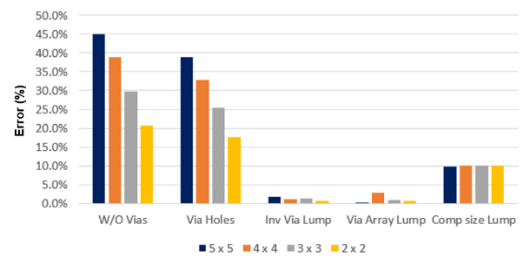


Chart 1: Simulated Temperature Error Against "Exact Vias Model"

3.2 Forced Cooling Simulation Analysis

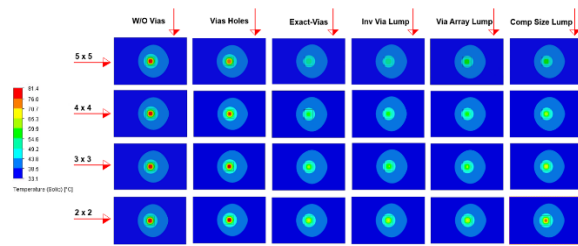


Figure 8: Temperature Contours of all modelling approaches

Array Size	Modelling Approach					
	W/O Vias	Via Holes	Exact Vias	Inv Array Lump	Via Array Lump	Comp Size Lump
	Component Temperature (°C)					
5 x 5	81.4	75.8	53.6	53.2	54.1	60.0
4 x 4		78.2	56.6	55.9	56.9	63.1
3 x 3		80.1	61.1	60.7	60.5	68.1
2 x 2		80.6	66.7	63.8	66.5	73.6

Table 3: Simulated Component Temperatures of all modelling approaches

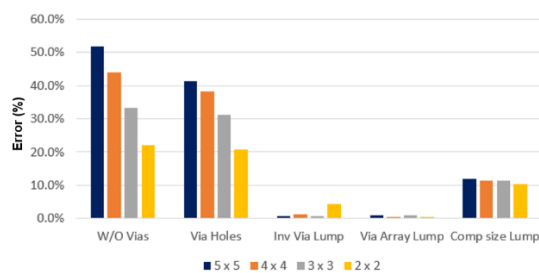


Chart 2: Simulated Temperature Error Against "Exact Vias Model"

4. Conclusion

It can be noted from the above simulation studies for both natural & forced cooling methods with different via array sizes that the "Lumped Individual Vias Model" & "Lumped Vias Array Model" consistently showed good agreements with "Exact Vias Model", not only the component temperature prediction & also the thermal images are closely matched.

It can be concluded based on the simulation studies conducted in this article that as "Lumped Vias Array Model" is quicker to model than other approaches discussed and it is also computationally efficient, this model can be well employed for Thermal Vias Modelling.

5. References

1. Nadezhda Kafadarova and Anna Andonova "PCB Thermal Design Improvement Through Thermal Vias", Department of Microelectronics, Technical University of Sofia, Bulgaria.
2. S. Lee, T.F. Lemczyk and M.M. Yovanovich, "Analysis of Thermal Vias in High Density Interconnect Technology", Microelectronics Heat Transfer Laboratory, Department of Mechanical Engineering, University of Waterloo, Ontario, Canada.